

# Economic Rationality and Substance Use Disorder

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## Abstract

We assess the decision-making quality of individuals with Substance Use Disorder (SUD) across risk, time, and social domains using measures of economic rationality from revealed preference theory. Relative to a control group, we find that the SUD cohort overall exhibits significantly lower CCEI scores in the risk and time domains, suggesting worse decision-making. Parametric preference estimates also reveal that individuals with SUD display elevated risk aversion and greater impatience. No significant differences in rationality or preferences are found in the social domain. We conclude by discussing potential implications of these findings for future research in modelling addiction and the use of economic rationality in policies targetting SUD relapse.

## 1 Introduction

The rational addiction model (Becker and Murphy, 1988) posits that addictive behavior can stem from utility-maximizing, forward-looking choices, implying that addicted individuals can maintain choice consistency. Other models accommodate addictive behaviors through specific forms of bounded rationality, such as time-inconsistent preferences or impulsive decision processes (Bernheim and Rangel, 2004). Critically, individuals modeled under such frameworks are assumed to possess the capabilities for consistent decision-making. However, from the perspective of neuroscience and psychology, evidence suggests that Substance Use Disorder (SUD) introduces profound cognitive burdens that alter decision-making ability across multiple behavioral domains (Bechara et al., 2001; Goldstein and Volkow, 2011). For example, cognitive impairments are often associated with degraded executive control and compromised working memory. This introduces a tension between the strength of the assumptions of rationality in economic models and clinical observations of deficits in general decision-making in persons with SUD (pwSUD). We explore this tension by investigating whether persons with SUD exhibit lower levels of decision-making quality as measured by economic rationality.

While adherence to the Generalized Axiom of Revealed Preference (GARP) has been utilized as a proxy for decision-making quality in other contexts (Choi et al., 2014; Chen et al., 2023), its application to those with SUD remains unexplored. We address this by conducting an experiment comparing

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persons with SUD to a control group. We find that individuals with SUD exhibit significantly lower rationality scores in the risk and time domains, dropping by 0.13, or 0.85 standard deviations (SDs) and 0.09 (0.67 SDs) respectively. Furthermore, our structural estimates reveal substantial differences in underlying preferences. Individuals with SUD display greater impatience (a 0.43 reduction in the discount factor, or 0.95 SDs) and elevated risk aversion (a 0.46 increase in the CRRA coefficient, or 0.94 SDs).

Naturally, this paper also builds on and contributes to the empirical literature that seeks to estimate the preferences of pwSUD. Previous studies have primarily examined addictive behavior by documenting preference heterogeneity (across and within domains), such as elevated time discounting (Ida et al., 2011; MacKillop et al., 2011; Petry, 2001), altered risk attitudes (Labeaga, 1999; Reynolds et al., 2018), or degraded social decision-making and altruism (Bickel et al., 2012; Preller et al., 2014). Typically, the current literature relies on highly structured elicitation tasks, such as multiple price lists or simple surveys. While these methods can be used to elicit (or describe) preferences, they lack the theoretical structure and data richness to formally quantify the severity inconsistent decision-making.<sup>1</sup> Thus, by using the convex budget set design (Choi et al., 2007; Andreoni and Miller, 2002; Andreoni and Sprenger, 2012), we are able to simultaneously elicit preference parameters and compute measures of decision-making quality.

The remainder of the paper is organized as follows. Section 2 details the experimental design and methodological outline for estimating rationality and structural parameters. Section 3 outlines our hypotheses. Section 4 presents the empirical results, and Section 5 concludes with a discussion of implications for policy, and future research.

## 2 Experiment

### 2.1 Sample and Procedures

The experiments were conducted in Deyang City, Sichuan Province, China. We recruited a total of  $N = 95$  individuals with SUD from a local rehabilitation clinic **month, year** and  $N = 37$  control subjects from a nearby community **month, year**.<sup>2</sup> The SUD sample was randomly selected using patient identification numbers, while the control group was drawn from a random sample of households based on residential addresses.<sup>3</sup> As rehabilitation clinics in China are heavily regulated, we faced several logistical constraints that necessitated deviations from standard laboratory protocols. First, restrictions on outside equipment required us to use pen-and-paper formats rather than computerized interfaces.<sup>4</sup>

<sup>1</sup>For example, when using MPLs in the risk domain, it is common to disallow multiple switching as no standard utility model can rationalise such choices.

<sup>2</sup>See Online Appendix E for i) summary statistics and balance checks of demographic characteristics and Big Five personality traits, and ii) summary of survey responses specific to persons with SUD (e.g. age at first substance use, substance use frequency,... etc...

<sup>3</sup>The study protocol was approved by the Institutional Review Board of Chengdu Normal University (Approval No. 2022120; Date: 2022-12-23) and pre-registered at the Social Science Registry (Trial 19182).

<sup>4</sup>Using multiple computers inside the clinic would have required individual monitoring by staff, which was not logistically feasible. Moreover, the administration raised concerns that introducing electronic devices and their associated equipment (e.g., cables) could pose additional health and safety risks.

Second, due to clinical regulations, participants' choices could not be incentivized.<sup>5</sup> For consistency, the control group participated under identical experimental conditions.<sup>6</sup>

## 2.2 Design

We utilized a standard convex budget set design (Choi et al., 2007) where subjects allocated an endowment of tokens between two distinct accounts. Although choices were hypothetical, tokens were explicitly framed to participants as specific monetary values (Chinese Yuan) so that participants understood the exact monetary trade-offs. We varied the relative prices (the effective exchange rate between tokens in the two accounts) while keeping the total endowment constant across 25 budget sets for each domain. Following Choi et al. (2007), the budget sets were chosen such that at least one axis intercept was 50 or greater, but never exceeded 100. The same 25 budget sets were used within each domain, but the order of the budget sets was randomized within each domain. All subjects faced the same 75 budget sets with random ordering of the domain of choice, with 1 budget set per page. The accounts represented either two equally probable states of the world (risk domain), an immediate payout versus a 1-month delayed payout (time domain), or an allocation between the subject and an anonymous individual (social domain).<sup>7</sup>

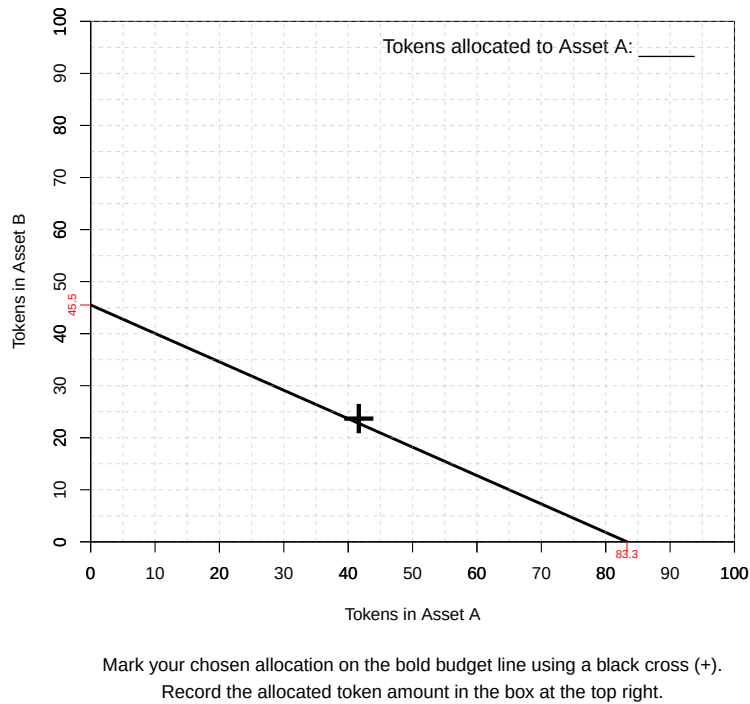


Figure 1: Example of a budget set round in the risk domain (translated from Mandarin Chinese)

<sup>5</sup>Although patients have access to personal bank accounts, the clinic explicitly prohibited financial incentives. This restriction was imposed due to safety concerns that the funds might be used inappropriately, and that disparities in experimental earnings could generate conflict among the patients.

<sup>6</sup>While experimental economics traditionally relies on incentivized choices, a robust literature demonstrates that hypothetical rewards often yield structural parameter estimates that are statistically indistinguishable from those elicited with real stakes, particularly in choices involving risk and intertemporal discounting (e.g., Camerer and Hogarth, 1999; Beattie and Loomes, 1997; Frederick et al., 2002).

<sup>7</sup>See Online Appendix A for experimental instructions and visual representations of the budget sets.

As illustrated in Figure 1, each decision presented subjects with a 2-D graph showing a downward sloping budget line. The axes displayed the maximum possible token allocation for each account (the intercepts), with the slope representing the relative price of allocating tokens to one account versus the other. Note that any choice along the 45-degree line represents an equal allocation of tokens to both accounts. As the experiment was conducted via pen-and-paper, subjects were explicitly instructed that their preferred allocation must lie on the budget line, and they were asked to mark this point and write down their chosen token amounts for Asset A. To ensure comprehension, an example was given for each domain, and subjects were asked if they understood the example before proceeding. Trained research assistants were available at all times to explain any questions related to any of the tasks.

### 3 Methodology

To evaluate economic rationality, we test for violations of GARP. We first define  $e$ -GARP. For a given efficiency level  $e \in (0, 1]$ , let a bundle  $\mathbf{q}_i$  be directly revealed preferred to bundle  $\mathbf{q}_j$  at efficiency level  $e$  (denoted  $\mathbf{q}_i R_e^0 \mathbf{q}_j$ ) if  $e\mathbf{p}_i \cdot \mathbf{q}_i \geq \mathbf{p}_j \cdot \mathbf{q}_j$ . The relation  $R_e$  is the transitive closure of  $R_e^0$  (indirect preference relation), meaning there exists a sequence of bundles  $\mathbf{q}_k, \dots, \mathbf{q}_m$  such that  $\mathbf{q}_i R_e^0 \mathbf{q}_k, \dots, \mathbf{q}_m R_e^0 \mathbf{q}_j$ . A dataset satisfies  $e$ -GARP if  $\mathbf{q}_i R_e \mathbf{q}_j$  implies  $e\mathbf{p}_j \cdot \mathbf{q}_j \leq \mathbf{p}_j \cdot \mathbf{q}_i$ . When  $e = 1$ , the data fully satisfies standard GARP, i.e., adherence to 1-GARP indicates that a subject’s choices are completely consistent with the maximization of a well-behaved utility function (Afriat, 1967; Varian, 1982). To quantify the severity of violations, we compute the Critical Cost Efficiency Index (CCEI). The CCEI is defined as the largest value of  $e$  such that the data satisfies  $e$ -GARP. Intuitively, a CCEI of  $e$  means the subject could have achieved the exact same level of utility by spending a fraction  $e$  of their budget efficiently. Thus,  $1 - e$  represents the proportion of wealth that the subject effectively “wasted” due to their inconsistent choices. The CCEI serves as our primary measure of decision-making quality (Choi et al., 2014).<sup>8</sup>

In addition to the standard CCEI, we also examine the CCEI-FOSD and the frequency of First-Order Stochastic Dominance (FOSD) compliance in the risk domain (Choi et al., 2014). For the CCEI-FOSD, we combine the actual choices made by a subject with the mirror image of those choices.<sup>9</sup> The CCEI-FOSD is the standard CCEI computed on this combined dataset of 50 choices. This measure effectively penalizes asymmetric allocations across identical states, as allocating less to the cheaper account is clearly a violation of FOSD, and generates a direct violation of GARP when evaluated against its mirror image. We also calculate the frequency of FOSD compliance out of the possible 25 decisions in the risk domain. Simply put, a choice violates FOSD if there exists another feasible allocation on the budget line that provides a higher expected return.<sup>10</sup>

To recover underlying structural preference parameters, we estimate utility functions for each domain using non-linear least squares based on theoretical expenditure shares derived from the first-order conditions. For risk preferences, suppose subjects allocate tokens between two equally probable states

<sup>8</sup>We repeat the analysis using other common measures of economic rationality, namely the Houtman-Maks Index (HMI), the Average Varian Index (AVI), and the Minimum Cost Index (MCI) in Online Appendix F.

<sup>9</sup>If a subject chooses allocation  $(x_1, x_2)$  with budget  $p_1x_1 + p_2x_2 = m$ , we assume they would have chosen  $(x_2, x_1)$  under the mirror-image budget constraint  $p_2x_1 + p_1x_2 = m$ .

<sup>10</sup>For example, if a unit of Asset A costs \$2 and a unit in Asset B costs \$1, choosing to allocate all your endowment to Asset 1 yields half as much as allocating it all to account 2. Since both accounts are equally likely to be chosen, the latter strictly stochastically dominates the former.

of the world  $(x_A, x_B)$ . We denote  $x_{max} = \max\{x_A, x_B\}$  as the high outcome and  $x_{min} = \min\{x_A, x_B\}$  as the low outcome. We specify the underlying utility function as:

$$U(x_A, x_B) = \alpha_r u(x_{max}) + (1 - \alpha_r) u(x_{min}) \quad (1)$$

where  $u(z) = z^{1-\rho_r}/(1 - \rho_r)$  for  $\rho_r \neq 1$  and  $u(z) = \ln(z)$  for  $\rho_r = 1$ . The parameter  $\alpha_r \in [0, 1]$  captures the decision weight placed on the better outcome (Gul, 1991).<sup>11</sup> When  $\alpha_r = 0.5$ , we have a standard expected utility (EU) function. When  $\alpha_r < 0.5$ , the better outcome is under-weighted relative to the objective probability of 0.5, implying disappointment aversion. The parameter  $\rho_r$  captures the coefficient of relative risk aversion (CRRA).

For time preferences, suppose the subject chooses the payment schedule  $(x_{today}, x_{later})$ . Following Andreoni and Sprenger (2012), we estimate the time discount factor  $\delta$  and curvature  $\rho_t$  assuming an additively separable utility function:

$$U(x_{today}, x_{later}) = u(x_{today}) + \delta u(x_{later}) \quad (2)$$

where  $u(\cdot)$  is defined similarly, and  $\rho_t$  represents the intertemporal curvature of the utility function. The discount factor  $\delta$  measures patience, where  $\delta < 1$  indicates standard discounting of future payoffs. The parameter  $\rho_t$  governs the desire to smooth payments across periods. A higher  $\rho_t$  corresponds to a stronger preference for consumption smoothing, while a lower  $\rho_t$  indicates a greater willingness to concentrate consumption in a single period.

For social preferences, subjects allocate tokens between themselves ( $x_s$ ) and an anonymous peer ( $x_o$ ) in a dictator game variant. Following Fisman et al. (2007), we specify the utility function as a member of the CES family:

$$U(x_s, x_o) = [\alpha_s x_s^{\rho_s} + (1 - \alpha_s) x_o^{\rho_s}]^{\frac{1}{\rho_s}} \quad (3)$$

The parameter  $\alpha_s \in [0, 1]$  indicates the weight placed on the personal payoff relative to the other's payoff, where  $\alpha_s = 0.5$  represents fair-mindedness and  $\alpha_s \rightarrow 1.0$  implies absolute self-regard. The parameter  $\rho_s$  represents the curvature of the indifference curves, measuring equality-efficiency orientation. When  $\rho_s = 1$ , the payments are perfect substitutes and the decision maker is fully efficiency-oriented. When  $\rho_s \rightarrow 0$ , the utility function approaches the Cobb-Douglas form meaning shares of expenditures are constant. As  $\rho_s \rightarrow -\infty$ , the function approaches Leontief utility, indicating a purely equality-oriented decision maker.<sup>12</sup>

<sup>11</sup>This formulation is equivalent to the disappointment aversion specification in Choi et al. (2007), who maximize  $\min\{\alpha_{AER}u(x_A) + u(x_B), u(x_A) + \alpha_{AER}u(x_B)\}$  for  $\alpha_{AER} > 1$ . Because the minimum always applies the larger weight  $\alpha_{AER}$  to the smaller payoff, we can drop the min operator by explicitly ordering the outcomes into  $x_{max}$  and  $x_{min}$ , yielding  $u(x_{max}) + \alpha_{AER}u(x_{min})$ . Normalizing the weights to sum to one, our weight on  $x_{max}$  becomes  $\alpha_r = 1/(1 + \alpha_{AER})$ . Thus,  $\alpha_r < 0.5$  corresponds to disappointment aversion ( $\alpha_{AER} > 1$ ).

<sup>12</sup>Additional details on the derivations of the expenditure shares and the estimation procedure are provided in Online Appendix G.

## 4 Hypotheses

As addiction is often associated with reduced working memory, heightened impulsivity, and degraded executive control (Bickel et al., 2014, 2012), addicted individuals may struggle to maintain stable preferences when faced with economic trade-offs. Research also indicates that impaired decision-making (e.g., impaired error monitoring) (Euser et al., 2013; Luijten et al., 2014) and certain neurobehavioral deficits (e.g., deficits in emotional empathy and theory of mind) (Quednow et al., 2017) are predictive of the onset of SUD. We hypothesize that these (neuro)cognitive burdens manifest as increased GARP and FOSD violations.

- **H1 - Rationality**

- **H1.1 (Risk Rationality):**  $CCEI_{\text{Risk, SUD}} < CCEI_{\text{Risk, Control}}$
- **H1.2 (Time Rationality):**  $CCEI_{\text{Time, SUD}} < CCEI_{\text{Time, Control}}$
- **H1.3 (Social Rationality):**  $CCEI_{\text{Social, SUD}} < CCEI_{\text{Social, Control}}$
- **H1.4 (Risk and FOSD Rationality):**  $CCEI\text{-}FOSD_{\text{SUD}} < CCEI\text{-}FOSD_{\text{Control}}$
- **H1.5 (FOSD):** Proportion of FOSD compliance in the risk domain is lower for persons with SUD

Regarding structural preferences, for the risk domain, individuals with SUD are generally characterized by elevated thrill-seeking and risk-taking behaviors when it comes to drug access (Lejuez et al., 2002; Takahashi et al., 2007), thus it seems natural to extend this argument to lotteries, anticipating greater risk-seeking behavior. Behavioral literature demonstrates severe myopia among addicted individuals (MacKillop et al., 2011), because drug use can shift intertemporal trade-offs toward immediate gratification. In the time domain, we anticipate this impulsivity to manifest as reduced patience and a lower desire to smooth consumption. In the social domain, SUD often socially isolates individuals and strains relationships, blunting their prosocial preferences (Preller et al., 2014). Furthermore, this isolation may lead them to prioritize strict equity (Rawlsian equality) over maximizing total social output (efficiency) when evaluating allocations.

- **H2 - Preferences**

- **H2.1 (Risk):** Persons with SUD are more risk-seeking ( $\rho_{r,\text{SUD}} < \rho_{r,\text{Control}}$ ).
- **H2.2A (Time):** Persons with SUD are less patient ( $\delta_{\text{SUD}} < \delta_{\text{Control}}$ ).
- **H2.2B (Time):** Persons with SUD exhibit less desire to smooth consumption ( $\rho_{t,\text{SUD}} < \rho_{t,\text{Control}}$ ).
- **H2.3A (Social):** Persons with SUD are more self-regarding ( $\alpha_{s,\text{SUD}} > \alpha_{s,\text{Control}}$ ).
- **H2.3B (Social):** Persons with SUD have a stronger preference for equity over efficiency ( $\rho_{s,\text{SUD}} < \rho_{s,\text{Control}}$ ).

To account for multiple hypothesis testing within each of the main hypotheses (H1 and H2), the subsequent regression tables report adjusted  $p$ -values using the Romano-Wolf step-down procedure (Romano and Wolf, 2005a,b). A detailed explanation of this procedure is provided in Online Appendix C.

## 5 Results

**Economic rationality:** The first three panels of Figure 2 show box plots for the CCEI in each choice domain. Visually, the CCEI distributions for individuals with SUD are shifted downward in both the risk and time domains, while the distributions in the social domain appear largely similar.

Tables 1 presents the main regression results for rationality, controlling for age, gender, education, income, and Big Five personality traits.<sup>13</sup> Confirming the visual patterns statistically, as shown in Panel A of Table 1, individuals with SUD exhibit significantly reduced CCEI in the risk and time domains relative to the control group, providing strong support for hypotheses H1.1 and H1.2. However, we find no significant difference in the social domain CCEI, meaning H1.3 is not supported. This suggests that while individuals with SUD have lower decision-making quality in domains involving personal monetary risk and intertemporal trade-offs, persons with SUD are still able to make consistent choices in the social domain. As a robustness check, Panel B contains equivalent regressions with a tobit specification, with similar conclusions.

From the two panels on the right of Figure 2, we also see that individuals with SUD also exhibit significantly lower scores on the FOSD measures in the risk domain (both CCEI-FOSD and raw FOSD compliance frequency). From Tables 1, although this is true statistically without considering multiple hypotheses corrections, the p-values adjusted for multiple hypotheses suggest only the CCEI-FOSD is statistically significant, hence H1.4 is supported with H1.5 only supported in isolation.

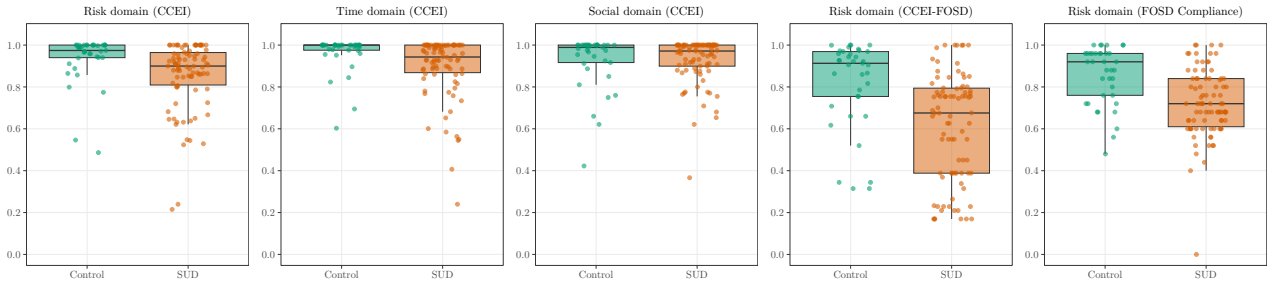


Figure 2: Comparison of Critical Cost Efficiency Index (CCEI) between SUD and Control Groups across Risk, Time, and Social domains.

<sup>13</sup>See Online Appendix E for summary statistics of these variables and Online Appendix H for full regression outputs. We also compute results using Houtman-Maks Index (HMI), the Average Variance Index (AVI), and the Minimum Cost Index (MCI) in Online Appendix F.

Table 1: Main Paper: CCEI, CCEI-FOSD, and FOSD Compliance

| Dependent Variable                           | (i) Without Controls |          |            |          | (ii) With Controls |          |            |          |
|----------------------------------------------|----------------------|----------|------------|----------|--------------------|----------|------------|----------|
|                                              | Coeff.               | SE       | Unadj. $p$ | FWER $p$ | Coeff.             | SE       | Unadj. $p$ | FWER $p$ |
| <b>Panel A: Ordinary Least Squares (OLS)</b> |                      |          |            |          |                    |          |            |          |
| CCEI (Risk)                                  | -0.0798**            | (0.0251) | 0.004      | 0.013    | -0.1308***         | (0.0395) | 0.002      | 0.008    |
| CCEI (Time)                                  | -0.0704***           | (0.0209) | 0.003      | 0.010    | -0.0911**          | (0.0385) | 0.035      | 0.050    |
| CCEI (Social)                                | 0.0045               | (0.0241) | 0.861      | 0.861    | -0.0059            | (0.0307) | 0.865      | 0.865    |
| CCEI-FOSD (Risk)                             | -0.2073***           | (0.0432) | 0.001      | 0.001    | -0.2217***         | (0.0595) | 0.004      | 0.008    |
| FOSD Compliance (Risk)                       | -0.1380              | (0.0283) | 0.001      | 0.103    | -0.1580            | (0.0421) | 0.004      | 0.191    |
| <b>Panel B: Tobit Regressions</b>            |                      |          |            |          |                    |          |            |          |
| CCEI (Risk)                                  | -0.1080***           | (0.0350) | 0.003      | 0.007    | -0.1660***         | (0.0471) | 0.001      | 0.008    |
| CCEI (Time)                                  | -0.1252***           | (0.0388) | 0.005      | 0.007    | -0.1668**          | (0.0600) | 0.031      | 0.037    |
| CCEI (Social)                                | 0.0023               | (0.0375) | 0.949      | 0.949    | -0.0142            | (0.0452) | 0.822      | 0.822    |
| CCEI-FOSD (Risk)                             | -0.2077***           | (0.0441) | 0.001      | 0.001    | -0.2213***         | (0.0556) | 0.004      | 0.008    |
| FOSD Compliance (Risk)                       | -0.1507              | (0.0322) | 0.001      | 0.108    | -0.1693            | (0.0427) | 0.005      | 0.307    |

Notes: Table reports regression coefficients for the Substance Use Disorder (SUD) indicator (SUD = 1, Control = 0). Conditional specifications control for gender, education, income, age, and Big Five personality traits. The sample consists of  $N = 129$  participants for OLS models and  $N = 129$  participants for Tobit regressions. Robust standard errors are reported in parentheses for OLS models; standard errors for Tobit regressions. FWER-adjusted  $p$ -values are computed via Romano-Wolf stepdown resampling with 1,000 permutations over the variables in each panel. Significance stars are based on FWER-adjusted  $p$ -values: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

**Structural preferences:** Figure 3 presents box plots for the estimated structural parameters. Across the entire sample, the risk weight ( $\alpha_r$ ) is clustered around 0.5, suggesting that subjects largely conform to expected utility. The risk curvature parameter ( $\rho_r$ ) indicates that the majority of subjects are risk-averse, while the time curvature parameter ( $\rho_t$ ) indicates a preference for consumption smoothing. In the time domain, subjects are predominantly standard time discounters ( $\delta < 1$ ),<sup>14</sup> and in the social domain, the altruism weights ( $\alpha_s$ ) indicate that subjects are largely self-regarding, while their elasticity parameters ( $\rho_s$ ) are distributed around zero, suggesting a balance between equity and efficiency. As a handful of subjects exhibited extreme choice behaviors leading to boundary-hitting parameter estimates, we exclude these extreme outliers from the figure and the OLS regressions to prevent them from driving the mean effects.<sup>15</sup> Table 2 presents the corresponding regression analysis.

**Risk Domain:** We find no robust difference in the risk weight ( $\alpha_r$ ) between the SUD and control groups suggesting subjects conforming with EU theory. However, our results contradict H2.1; rather than being risk-seeking, SUD subjects exhibit significantly greater CRRA risk aversion ( $\rho_r$ ). While this effect is statistically significant at the 5% level across all models in isolation, we note that it is attenuated after multiple hypotheses correction losing statistical significance in the median specifications. A plausible explanation for elevated risk aversion is that we incorrectly assumed that the risk-seeking behavior associated with substance use (and all accompanying activities) would transfer to the financial domain. Indeed, Weber et al. (2002) demonstrate that risk preferences are highly domain-specific rather than a stable trait. In the context of SUD, financial decision-making may reflect a more cautious approach to risk in order to sustain substance use. This idea is consistent with economic research demonstrating

<sup>14</sup>Interestingly, although the majority of subjects are standard time discounters ( $\delta < 1$ ), a handful of subjects exhibit extreme patience ( $\delta > 1$ ). While uncommon, this accurately reflects systematic majority allocation of wealth to the later period across intertemporal tasks, which can only be explained by a stronger preference for future consumption.

<sup>15</sup>Specifically, we trim the top 5% of estimates for  $\rho_r$ ,  $\rho_t$ , and  $\delta$ , and the bottom 5% for  $\rho_s$ . To ensure we still take the full sample into account, we also estimate median regressions on the untrimmed sample. Full un-trimmed box plots are available in the Online Appendix.

that poverty and financial scarcity induce elevated risk aversion ([Haushofer and Fehr, 2014](#)).

**Social Domain:** Finally, we find no statistically significant difference in either the altruism weight ( $\alpha_s$ ) or the altruistic substitutability ( $\rho_s$ ) between the SUD and control groups. Thus, neither H2.3A nor H2.3B is supported, mirroring the lack of difference seen in the social CCEI. While clinical observations frequently link SUD with antisocial-type behavior, these tendencies are heavily state-dependent, driven by acute visceral contexts rather than a fundamental shift in baseline social preferences. This is in contrast to the risk and time domains, where clinical and behavioral observations are more persistent, not necessarily requiring a visceral state to manifest.<sup>16</sup> Consequently, when evaluated in an (abstract) experimental setting without such immediate pressures, individuals with SUD appear to maintain their baseline social preference.

**Time Domain:** We find strong support for H2.2A, as SUD subjects exhibit a significantly lower discount factor ( $\delta$ ), reflecting less patience. This effect is highly robust across both OLS and median regression specifications. Contrary to H2.2B, SUD subjects display a significantly higher intertemporal curvature ( $\rho_t$ ), indicating a stronger preference for consumption smoothing. This effect is also robust across most specifications. This elicited preference for consumption smoothing is likely driven by the physical necessities imposed by SUD. Within the framework of [Becker and Murphy \(1988\)](#), the consumption of an addictive good increases the need for it in subsequent periods. This reinforcing nature implies that individuals strive to maintain a steady level of consumption to avoid the severe negative utility of withdrawal. As such, individuals with SUD may strongly prefer a steady and predictable flow of financial resources to reliably sustain their habits, which may drive a heightened aversion to intertemporal variation.

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<sup>16</sup>For instance, intense visceral states such as active craving or withdrawal can temporarily override long-term prosocial goals ([Loewenstein, 1996](#)).

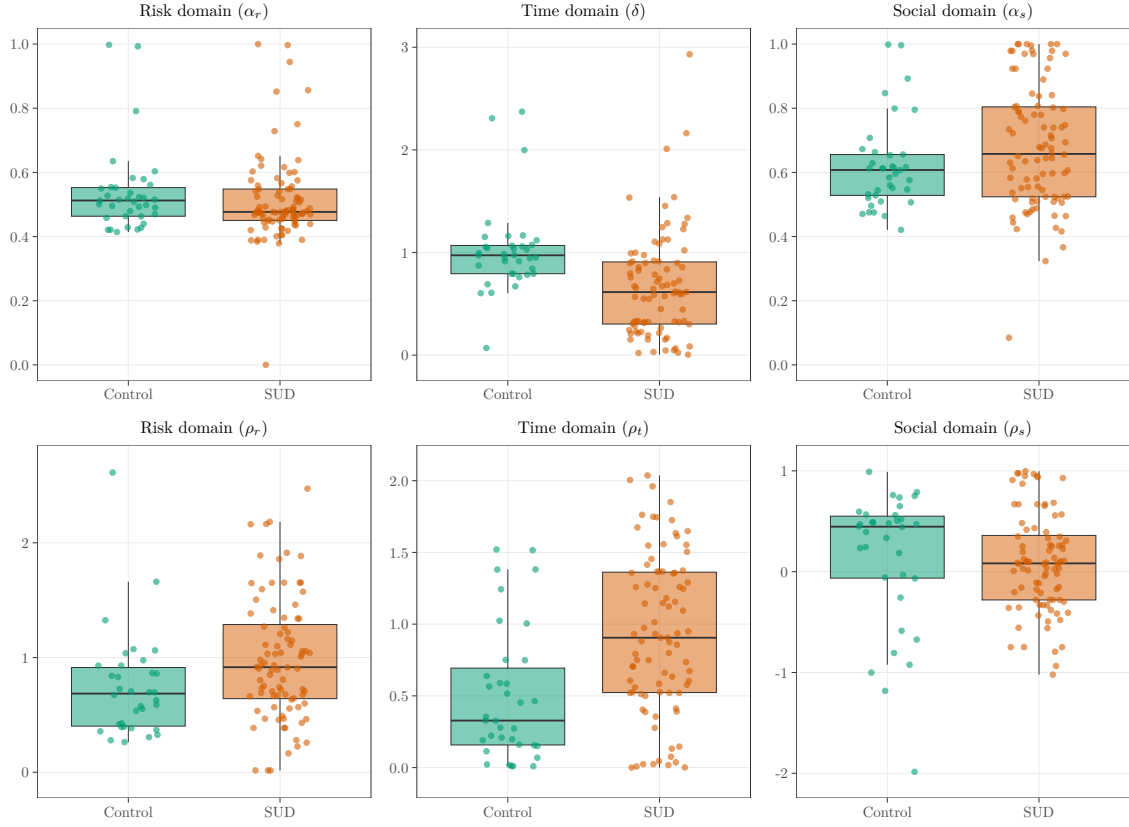


Figure 3: Comparison of Structural Preference Parameters between SUD and Control Groups, excluding extreme outliers.

Table 2: Main Paper: Primary Structural Parameters

| Dependent Variable                           | (i) Without Controls |          |            |          | (ii) With Controls |          |            |          |
|----------------------------------------------|----------------------|----------|------------|----------|--------------------|----------|------------|----------|
|                                              | Coeff.               | SE       | Unadj. $p$ | FWER $p$ | Coeff.             | SE       | Unadj. $p$ | FWER $p$ |
| <b>Panel A: Ordinary Least Squares (OLS)</b> |                      |          |            |          |                    |          |            |          |
| Risk Weight ( $\alpha_r$ )                   | -0.0243              | (0.0258) | 0.364      | 0.570    | 0.0207             | (0.0372) | 0.595      | 0.770    |
| CRRA Risk Aversion ( $\rho_r$ )              | 0.2381*              | (0.0966) | 0.021      | 0.080    | 0.3159*            | (0.1332) | 0.015      | 0.077    |
| Time Discount Factor ( $\delta$ )            | -0.3511***           | (0.0901) | 0.001      | 0.003    | -0.4917***         | (0.1049) | 0.001      | 0.002    |
| Time CRRA ( $\rho_t$ )                       | 0.4324***            | (0.0976) | 0.001      | 0.002    | 0.4130**           | (0.1248) | 0.004      | 0.015    |
| Social Weight ( $\alpha_s$ )                 | 0.0588               | (0.0305) | 0.075      | 0.197    | 0.0650             | (0.0421) | 0.125      | 0.328    |
| Social CES ( $\rho_s$ )                      | -0.0453              | (0.1271) | 0.710      | 0.710    | 0.0957             | (0.1480) | 0.515      | 0.770    |
| <b>Panel B: Median Regressions</b>           |                      |          |            |          |                    |          |            |          |
| Risk Weight ( $\alpha_r$ )                   | -0.0356**            | (0.0099) | 0.015      | 0.022    | -0.0074            | (0.0329) | 0.817      | 0.817    |
| CRRA Risk Aversion ( $\rho_r$ )              | 0.2331               | (0.1083) | 0.023      | 0.127    | 0.4004             | (0.1793) | 0.047      | 0.159    |
| Time Discount Factor ( $\delta$ )            | -0.3187**            | (0.0687) | 0.002      | 0.016    | -0.5028**          | (0.1330) | 0.002      | 0.012    |
| Time CRRA ( $\rho_t$ )                       | 0.5778**             | (0.1560) | 0.001      | 0.016    | 0.4956             | (0.2531) | 0.053      | 0.182    |
| Social Weight ( $\alpha_s$ )                 | 0.0479               | (0.0416) | 0.183      | 0.183    | 0.1142             | (0.0618) | 0.070      | 0.182    |
| Social CES ( $\rho_s$ )                      | -0.3159              | (0.1238) | 0.085      | 0.127    | -0.1286            | (0.2498) | 0.493      | 0.779    |

Notes: Table reports regression coefficients for the Substance Use Disorder (SUD) indicator (SUD = 1, Control = 0). Conditional specifications control for gender, education, income, age, and Big Five personality traits. The sample consists of  $N = 129$  participants for OLS models and  $N = 129$  participants for Median regressions. Robust standard errors are reported in parentheses for OLS models; bootstrap standard errors (100 replications) for Median regressions. FWER-adjusted  $p$ -values are computed via Romano-Wolf stepdown resampling with 1,000 permutations over the variables in each panel. Significance stars are based on FWER-adjusted  $p$ -values: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

**Additional results 1 (power analysis):** While we find lower decision-making quality in persons

with SUD, it is important to emphasize that their choices are still more consistent than random behavior. This speaks to the power of GARP as a criterion for decision-making, as well as a robustness check for whether individuals just chose randomly to complete the tasks. Following [Dean and Martin \(2016\)](#), we evaluate this absolute rationality by comparing subjects against 1,000 simulated uniform random choosers on the exact same budget sets. The resulting Selten score measures how much the observed CCEIs exceeds the average CCEI of the random choosers, while the percentile score ranks them against the simulated choosers. A positive Selten score indicates choices are being made according to preferences, rather than chance. As shown in Table 3, both SUD and control subjects have positive Selten indices and high percentile scores across all domains confirming that GARP as a test is sufficiently powerful.

Table 3: Average Rationality Measures by Domain and Group

| Group   | Risk  |        |      | Time  |        |      | Social |        |      |
|---------|-------|--------|------|-------|--------|------|--------|--------|------|
|         | CCEI  | Selten | Pct. | CCEI  | Selten | Pct. | CCEI   | Selten | Pct. |
| SUD     | 0.855 | 0.132  | 78.9 | 0.896 | 0.173  | 85.5 | 0.926  | 0.204  | 90.4 |
| Control | 0.935 | 0.212  | 91.8 | 0.962 | 0.240  | 94.8 | 0.927  | 0.204  | 89.0 |

*Notes:* This table presents the average CCEI, Selten score, and Percentile score (Pct.) against uniformly random choice for both the SUD and Control groups across the risk, time, and social domains. Following [Dean and Martin \(2016\)](#), the Selten score is defined as a subject’s actual CCEI minus the mean CCEI across 1,000 simulated uniform random choices on the same budget sets. The percentile score (Pct.) measures the percentage of simulated random choices whose CCEI score is less than or equal to the subject’s actual score.

**Additional results 2 (relapse):** A natural question is whether lower decision-making quality, as proxied by economic rationality, has useful clinical implications such as predicting relapse frequency. To examine this, we analyze the frequency of admissions at a rehabilitation clinic, which is measured in 3 categories (‘1 time’, ‘2-3 times’, and ‘4-5 times’) in Table 4 with Ordered Logit regressions. While no causal link can be asserted, decision-making quality in the time domain is negatively associated with higher relapse frequency. For example, in the model with controls (Column 8), a 1 standard deviation increase in the time CCEI decreases the log odds of being in a higher category of compulsory detox admissions (e.g., 2–3 times or 4–5 times vs. 1 time) by 1.809, *ceteris paribus*. This negative association is persistent across all specifications. Neither the risk- nor the social-domain displays any consistent relationship with relapse frequency. Overall, this can suggest that using revealed preference measures of intertemporal decision-making consistency may have potential use when analysing relapse vulnerability. If poor decision-making in the time domain is a fundamental driver of relapse frequency, interventions targeting this form of decision-making could prove useful in SUD treatment.

Table 4: Ordered Logit Regression of Relapse Frequency on Rationality Indices (1 SD Log Odds, SUD Sample Only)

|                     | (1)               | (2)                | (3)               | Relapse Frequency |                  | (6)                 | (7)               | (8)                  |
|---------------------|-------------------|--------------------|-------------------|-------------------|------------------|---------------------|-------------------|----------------------|
|                     |                   |                    |                   | (4)               | (5)              |                     |                   |                      |
| CCEI (Risk, 1 SD)   | −0.345<br>(0.239) |                    |                   | −0.340<br>(0.246) | 0.319<br>(0.494) |                     |                   | 0.456<br>(0.579)     |
| CCEI (Time, 1 SD)   |                   | −0.459*<br>(0.257) |                   | −0.506<br>(0.317) |                  | −0.952**<br>(0.416) |                   | −1.809***<br>(0.686) |
| CCEI (Social, 1 SD) |                   |                    | −0.167<br>(0.235) | 0.092<br>(0.294)  |                  |                     | −0.125<br>(0.337) | 1.045*<br>(0.580)    |
| Controls            | No                | No                 | No                | No                | Yes              | Yes                 | Yes               | Yes                  |
| Observations        | 64                | 64                 | 64                | 64                | 53               | 53                  | 53                | 53                   |

Notes: Table reports Ordered Logit coefficients (log odds). The dependent variable is the reported category of compulsory detox admissions (1 = “1 time”, 2 = “2-3 times”, 3 = “4-5 times”). Predictors are standardized to have a mean of 0 and standard deviation of 1. In the estimation sample, 1 SD is 0.146 for CCEI (Risk), 0.115 for CCEI (Time), and 0.111 for CCEI (Social). Columns (1)-(4) show models without demographic controls, and Columns (5)-(8) include demographic and personality controls. Standard errors are reported in parentheses. Sample includes only SUD subjects. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

## 6 Concluding Remarks

In this paper, we document that SUD is associated with lower decision-making quality, as measured by economic rationality, specifically in the domains of risk and time. Furthermore, their elicited preferences are characterized by severe intertemporal impatience and heightened risk aversion. These results indicate a pattern of inconsistent decision-making that aligns with persons with SUD. Overall, our preliminary findings suggest that economic rationality provides a relevant and powerful (in the statistical sense) framework for analyzing SUD, offering a set of decision-making measures for understanding for relapse vulnerability. Policy interventions targeting executive function and decision-making stability could prove vital in future SUD treatment.

Our preliminary investigation into the decision-making quality of persons with SUD precludes making any causal claims. Establishing causality would require future research involving an incentivized, long-term, longitudinal field experiment in which randomized treatments actively target decision-making quality (across all domains) among persons with SUD, for example, by enhancing cognitive control through neurofeedback training. Additionally, our findings face external validity concerns, as the clinical and demographic characteristics of persons with SUD are often highly culture- and country-specific. Nevertheless, our results remain highly relevant; they successfully capture the behavioral phenotype of SUD within Sichuan (an incredibly populous province within China) and may be generalizable to the broader Chinese context. Despite these limitations, our study establishes that economic rationality is a useful and robust measure for incorporating behavioral economics into the future study of SUD.

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# Online Appendices

## A Experimental Design and Instructions

We utilized a sequential budget set design to test economic rationality. Participants faced a series of continuous linear budget constraints and allocated tokens between two distinct accounts. These accounts varied depending on the domain:

- **Risk Domain:** Accounts represented two mutually exclusive states of the world, each with a 50% probability of occurring.
- **Time Domain:** Accounts represented payouts at an earlier date (e.g., today) versus a later date (e.g., in 4 weeks).
- **Social Domain:** Accounts represented tokens kept by the participant versus tokens given to an anonymous peer.

By varying the relative prices and total income across 25 sequential trials per domain, we mapped each participant's underlying indifference curves. Because this experiment involved a clinical Substance Use Disorder population, participants were not monetarily incentivized. However, to ensure comprehension and tangible framing, the “tokens” in all budget set visual instruments were explicitly denominated in specific monetary values (Yuan) corresponding to real-world purchasing power. The tasks were completed in person using a pen-and-paper format.

### A.1 Participant Instructions across Behavioral Domains (English Translations)

Below, we provide the verbatim English translations of the experimental instructions provided to participants prior to each task across all three behavioral domains:

#### A.1.1 Domain 1: Risk Preferences (Risk / Uncertainty)

*“This questionnaire presents a series of economic decision-making scenarios. Please respond according to your true preferences. The questionnaire consists of 25 independent decision rounds. In each round, your task is identical: allocate tokens between Asset A and Asset B. You have a 50% probability of receiving the payout from Asset A, and a 50% probability of receiving the payout from Asset B. The line depicts all possible allocations of tokens between Asset A (horizontal axis, x-axis) and Asset B (vertical axis, y-axis). Please mark a black cross (+) on the bold budget line in each trial to indicate your chosen allocation, and write the allocated token amount in the box at the top right.”*

#### A.1.2 Domain 2: Time Preferences (Intertemporal Choice)

*“This questionnaire presents a series of economic decision-making scenarios. Please respond according to your true preferences. The questionnaire consists of 25 independent decision rounds.*

In each round, your task is to allocate tokens between a payout received today and a payout received 1 month later. The budget line depicts all possible allocations between money received today (horizontal axis, x-axis) and money received 1 month later (vertical axis, y-axis). Please mark a black cross (+) on the bold budget line in each trial to indicate your chosen allocation, and write the allocated token amount in the box at the top right.”

### A.1.3 Domain 3: Social Preferences (Dictator Game / Altruism)

“This questionnaire presents a series of economic decision-making scenarios. Please respond according to your true preferences. The questionnaire consists of 25 independent decision rounds. In each round, you are randomly paired with a new anonymous participant. Your task is to allocate tokens between yourself and the matched anonymous recipient. The budget line depicts all possible monetary allocations for yourself (horizontal axis, x-axis) and for the matched recipient (vertical axis, y-axis). Please mark a black cross (+) on the bold budget line in each trial to indicate your chosen allocation, and write the allocated token amount in the box at the top right.”

## A.2 Budget Set Task Examples (Original Chinese and English Translations)

Figures 4–6 provide visual examples of the budget lines presented to participants during the experiment in both the original Chinese format and the translated English version.

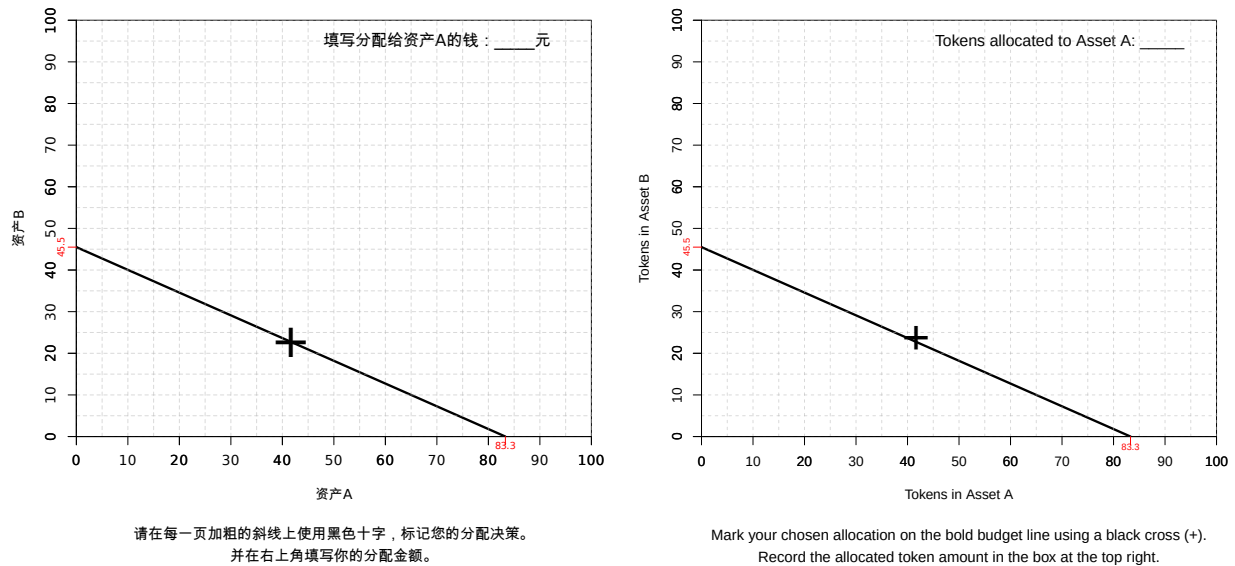
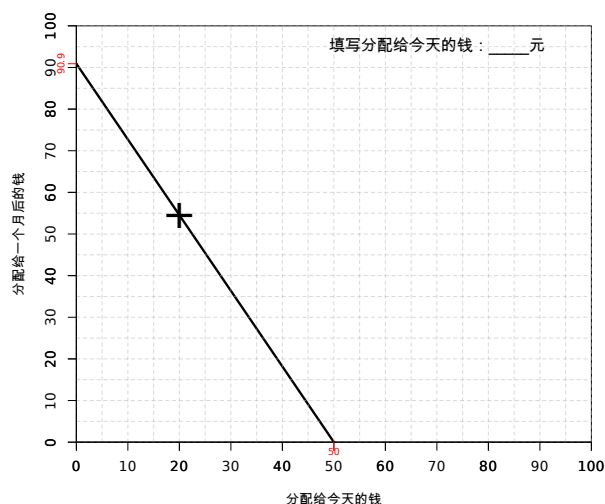
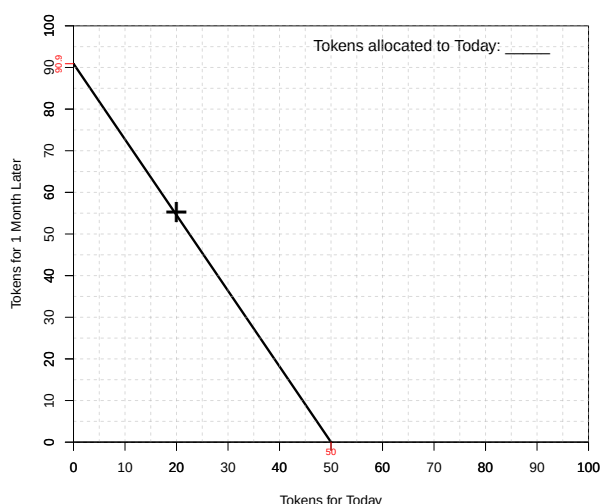


Figure 4: Example of a Risk Domain budget set task. Left: Original Chinese instrument presented to participants. Right: English translation. Participants mark a cross (+) on the budget line to allocate tokens between State 1 and State 2.

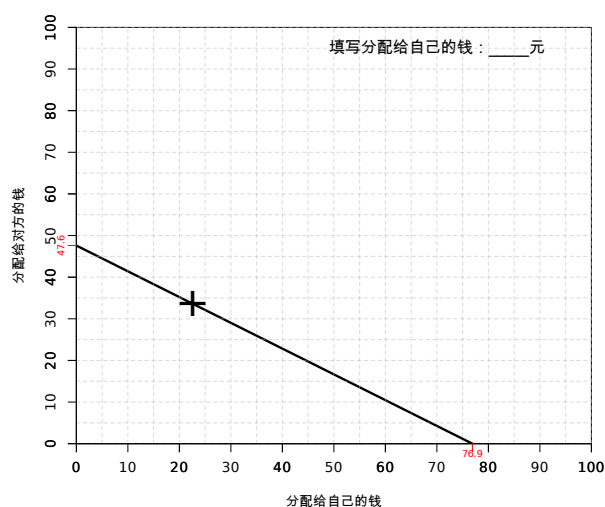


请在每一页加粗的斜线上使用黑色十字，标记您的分配决策。  
并在右上角填写你的分配金额。

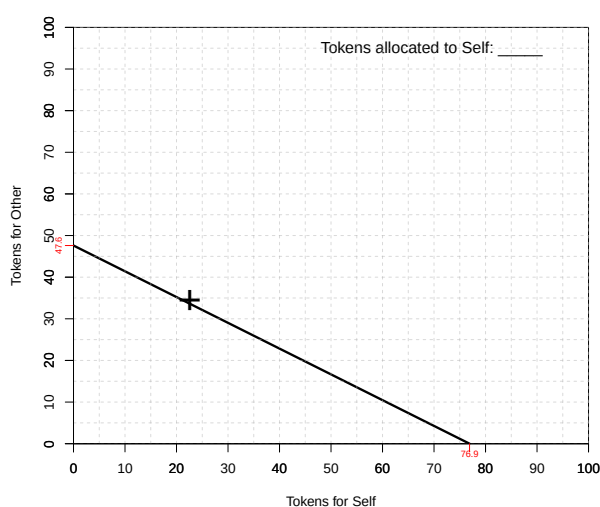


Mark your chosen allocation on the bold budget line using a black cross (+).  
Record the allocated token amount in the box at the top right.

Figure 5: Example of a Time Domain budget set task. Left: Original Chinese instrument. Right: English translation. Participants allocate tokens between an immediate payout today ( $t$ ) and a delayed payout ( $t + k$ ).



请在每一页加粗的斜线上使用黑色十字，标记您的分配决策。  
并在右上角填写你的分配金额。



Mark your chosen allocation on the bold budget line using a black cross (+).  
Record the allocated token amount in the box at the top right.

Figure 6: Example of a Social Domain budget set task. Left: Original Chinese instrument. Right: English translation. Participants allocate tokens between themselves and an anonymous recipient.

## B Structural Estimation Methodology

This section details the identification of structural utility parameters from choices over continuous linear budget sets. Following [Fisman et al. \(2007, 2015, 2017\)](#), we estimate the preference parameters using Non-Linear Least Squares (NLLS) by matching the observed budget expenditure shares with the theoretically optimal ones. The specific parameter derivations and NLLS objectives for each domain are outlined below.

**Risk Domain ( $\alpha_r$  and  $\rho_r$ ):** For this domain, we follow [Choi et al. \(2007\)](#). Subjects allocate endowments between two equally probable states. Let  $p_{cheap} = \min(p_A, p_B)$  and  $p_{exp} = \max(p_A, p_B)$ . We denote the allocation to the cheaper asset as  $x_{max} = \max(x_A, x_B)$  and the other as  $x_{min} = \min(x_A, x_B)$ . The utility function is:

$$U(x_A, x_B) = \alpha_r \frac{x_{max}^{1-\rho_r}}{1-\rho_r} + (1-\alpha_r) \frac{x_{min}^{1-\rho_r}}{1-\rho_r} \quad (4)$$

The optimality condition yields the tangency condition:

$$\left( \frac{x_{max}}{x_{min}} \right)^{-\rho_r} = \frac{1-\alpha_r}{\alpha_r} \frac{p_{cheap}}{p_{exp}} \quad (5)$$

Rearranging this condition allows us to express the theoretical optimal expenditure share allocated to the cheaper asset,  $s_{cheap}^* = \frac{p_{cheap} x_{max}}{m}$ :

$$s_{cheap}^*(p_{cheap}, p_{exp}) = \frac{g}{(p_{cheap}/p_{exp})^r + g} \quad \text{where} \quad r = \frac{1-\rho_r}{\rho_r}, \quad g = \left( \frac{\alpha_r}{1-\alpha_r} \right)^{\frac{1}{\rho_r}} \quad (6)$$

If Asset A is the cheaper asset, the theoretical expenditure share for Asset A is  $s_A^* = s_{cheap}^*$ ; otherwise,  $s_A^* = 1 - s_{cheap}^*$ . We estimate the parameters  $(\hat{\alpha}_{r,i}, \hat{\rho}_{r,i})$  for each subject  $i$  by minimizing the sum of squared differences between their observed expenditure share on Asset A,  $s_{A,i,t}$ , and the theoretical share  $s_A^*$ :

$$(\hat{\alpha}_{r,i}, \hat{\rho}_{r,i}) = \arg \min_{\alpha_r, \rho_r} \sum_{t=1}^{25} (s_{A,i,t} - s_A^*(\alpha_r, \rho_r; p_{A,i,t}, p_{B,i,t}))^2 \quad (7)$$

**Time Domain ( $\delta$  and  $\rho_t$ ):** For this domain, we follow [Andreoni and Sprenger \(2012\)](#). Subjects allocate tokens between an immediate payout ( $x_{today}$ ) and a delayed payout ( $x_{later}$ ). We posit an additively separable utility model:

$$U(x_{today}, x_{later}) = \frac{x_{today}^{1-\rho_t}}{1-\rho_t} + \delta \frac{x_{later}^{1-\rho_t}}{1-\rho_t} \quad (8)$$

The optimality condition yields the tangency condition:

$$\left( \frac{x_{today}}{x_{later}} \right)^{-\rho_t} = \delta \frac{p_{today}}{p_{later}} \quad (9)$$

Rearranging this isolates the theoretical optimal expenditure share allocated to the immediate payout,

$$s_{today}^* = \frac{p_{today}x_{today}}{m}.$$

$$s_{today}^*(p_{today}, p_{later}) = \frac{g}{(p_{today}/p_{later})^r + g} \quad \text{where} \quad r = \frac{1 - \rho_t}{\rho_t}, \quad g = \left(\frac{1}{\delta}\right)^{\frac{1}{\rho_t}} \quad (10)$$

We estimate the parameters  $(\hat{\delta}_i, \hat{\rho}_{t,i})$  by minimizing the sum of squared errors between the observed expenditure share for the immediate payout,  $s_{today,i,t}$ , and the theoretical share  $s_{today}^*$ :

$$(\hat{\delta}_i, \hat{\rho}_{t,i}) = \arg \min_{\delta, \rho_t} \sum_{t=1}^{25} \left( s_{today,i,t} - s_{today}^*(\delta, \rho_t; p_{today,i,t}, p_{later,i,t}) \right)^2 \quad (11)$$

**Social Domain ( $\alpha_s$  and  $\rho_s$ ):** For this domain, we follow [Fisman et al. \(2007\)](#). Subjects allocate tokens between themselves ( $x_{self}$ ) and an anonymous peer ( $x_{other}$ ). We model this via a CES function:

$$U(x_{self}, x_{other}) = \left[ \alpha_s x_{self}^{\rho_s} + (1 - \alpha_s) x_{other}^{\rho_s} \right]^{\frac{1}{\rho_s}} \quad (12)$$

Because any monotonic transformation preserves ordinal preferences, maximizing this utility is identical to maximizing  $\alpha_s \frac{x_{self}^{\rho_s}}{\rho_s} + (1 - \alpha_s) \frac{x_{other}^{\rho_s}}{\rho_s}$ . The optimality condition yields:

$$\left( \frac{x_{self}}{x_{other}} \right)^{\rho_s - 1} = \frac{1 - \alpha_s}{\alpha_s} \frac{p_{self}}{p_{other}} \quad (13)$$

Rearranging this isolates the theoretical optimal expenditure share for the personal payout,  $s_{self}^* = \frac{p_{self}x_{self}}{m}$ :

$$s_{self}^*(p_{self}, p_{other}) = \frac{g}{(p_{self}/p_{other})^r + g} \quad \text{where} \quad r = \frac{\rho_s}{1 - \rho_s}, \quad g = \left( \frac{\alpha_s}{1 - \alpha_s} \right)^{\frac{1}{1 - \rho_s}} \quad (14)$$

We estimate the parameters  $(\hat{\alpha}_{s,i}, \hat{\rho}_{s,i})$  by minimizing the sum of squared errors between the observed expenditure share for the personal payout,  $s_{self,i,t}$ , and the theoretical share  $s_{self}^*$ :

$$(\hat{\alpha}_{s,i}, \hat{\rho}_{s,i}) = \arg \min_{\alpha_s, \rho_s} \sum_{t=1}^{25} \left( s_{self,i,t} - s_{self}^*(\alpha_s, \rho_s; p_{self,i,t}, p_{other,i,t}) \right)^2 \quad (15)$$

Notice that the optimality conditions for each domain involves division by one of the choices, which can be an issue when we have corner solutions, hence why the matching of expenditure shares is the preferred estimation technique.

## C Multiple Hypothesis Testing: Romano-Wolf Step-Down Procedure

Because our analysis tests multiple interrelated hypotheses across three behavioral domains (risk, time, and social preferences) simultaneously, evaluating statistical significance using unadjusted (naïve)  $p$ -values increases the probability of false positives (Type I errors). To address this, we control the Family-Wise Error Rate (FWER) using the Romano-Wolf step-down multiple testing procedure (Romano and Wolf, 2005a,b).

We follow the approach of Kim et al. (2018), but we improve upon their methodology by employing the Romano-Wolf procedure rather than the Westfall-Young permutation procedure. The Westfall-Young method fundamentally relies on the assumption of subset pivotality, which implies that the joint distribution of test statistics for any subset of true null hypotheses is unaffected by the truth or falsehood of the remaining hypotheses. While subset pivotality generally holds in simple unadjusted comparisons (such as basic permutation tests between two groups), it frequently fails or is highly restrictive in the context of multivariate regression models with covariates. In contrast, the Romano-Wolf step-down procedure does not require the subset pivotality condition, rendering it asymptotically valid under much broader settings and making it the appropriate choice for our regression-based analysis.

In practice, the procedure works iteratively. First, the test statistics for all hypotheses in the family are calculated. A joint distribution of these statistics is simulated via bootstrap resampling (in our analysis, we use 1,000 bootstrap replications). The hypotheses are ordered from the most to the least significant (i.e., from largest to smallest test statistic). At each step, a hypothesis is rejected if its test statistic exceeds a critical value derived from the maximum test statistic in the remaining pool of hypotheses. Once a hypothesis fails to be rejected, the step-down process stops, and all subsequent hypotheses in the order are also retained.

For our OLS regression models, we calculate the adjusted  $p$ -values by bootstrapping the  $t$ -statistics of the SUD group indicator across our domain-specific outcomes (e.g., CCEI across the three domains). For the conditional median regressions (which can be highly sensitive to sparsity in small, resampled datasets), the bootstrapped estimates are studentized using the original sample's standard error to ensure mathematical stability and prevent test statistic explosion during resampling (Efron and Tibshirani, 1993). The resulting Romano-Wolf adjusted  $p$ -values provide a robust measure of significance that correctly reflects the multiplicity of our tests without sacrificing excessive statistical power.

## D Distribution of Structural Parameters and Outliers

Figure 7 presents the full, unfiltered distribution of the structural preference parameters. In our estimation procedure, we set wide parameter bounds of  $[-50, 50]$  to allow the Non-Linear Least Squares (NLLS) optimizer to converge freely without artificial truncation. As shown in the figure, one subject in the risk domain hit the upper limit ( $\rho_r = 50$ ), and one subject in the social domain hit the lower limit ( $\rho_s = -50$ ).

These boundary estimates are not algorithmic errors; rather, they accurately reflect highly extreme choice behaviors by these participants. For instance, the subject with an estimated  $\rho_r = 50$  (and  $\alpha_r \approx 0$ ) allocated 100% of their budget to the more expensive asset in all 25 decisions, forcing the optimizer to infer an extreme preference for minimizing their own payout. To ensure our statistical group comparisons in the main text are representative and not driven by these extreme phenotypic expressions, we systematically trim the top 5% of values for  $\rho_r$ ,  $\rho_t$ , and  $\delta$ , and the bottom 5% of values for  $\rho_s$ .

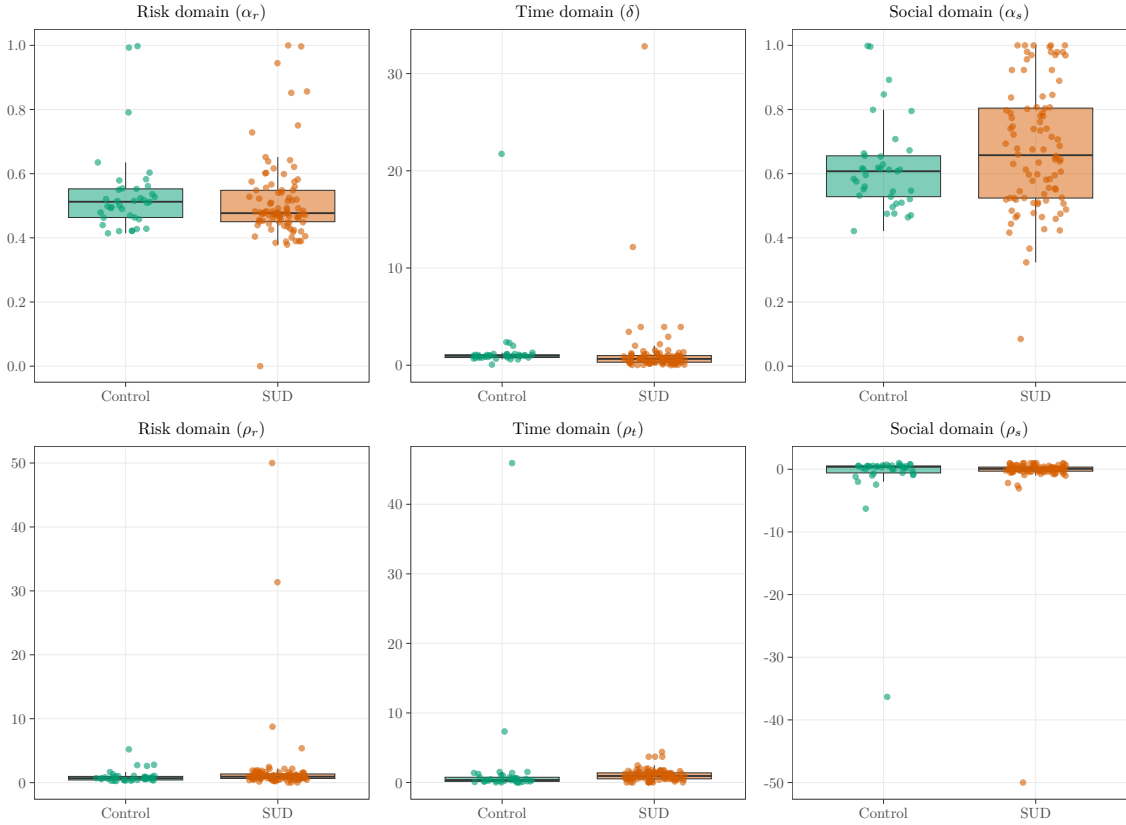


Figure 7: Unfiltered distribution of structural preference parameters across domains, showing extreme values hitting the estimation bounds of  $[-50, 50]$ .

## E Summary Statistics of Demographic and Personality Characteristics

To provide a comprehensive descriptive overview of the participant sample across groups, Table 5 presents the summary statistics for demographic characteristics and Big Five personality traits comparing the Substance Use Disorder (SUD) group and non-dependent Control group.

Table 5: Summary Statistics of Demographic and Personality Characteristics by Group

| Variable                           | SUD ( $N = 89$ ) | Control ( $N = 37$ ) | $p$ -value |
|------------------------------------|------------------|----------------------|------------|
| <b>Demographic Characteristics</b> |                  |                      |            |
| Age (Years)                        | 34.94 (9.59)     | 35.49 (10.42)        | 0.786      |
| <b>Gender</b>                      |                  |                      |            |
| Female                             | 35.96%           | 48.65%               | 0.199      |
| Male                               | 64.04%           | 51.35%               | 0.199      |
| <b>Education Level</b>             |                  |                      |            |
| High School or Below               | 13.48%           | 18.92%               | 0.470      |
| Bachelor's or College              | 70.79%           | 72.97%               | 0.806      |
| Master's+                          | 11.24%           | 8.11%                | 0.582      |
| Unspecified                        | 4.49%            | 0.00%                | 0.045**    |
| <b>Monthly Household Income</b>    |                  |                      |            |
| < 100,000 RMB                      | 61.80%           | 51.35%               | 0.291      |
| 100,000-500,000 RMB                | 14.61%           | 48.65%               | 0.000***   |
| > 500,000 RMB                      | 15.73%           | 0.00%                | 0.000***   |
| Unspecified                        | 7.87%            | 0.00%                | 0.007***   |
| <b>Big Five Personality Traits</b> |                  |                      |            |
| Extraversion                       | 3.34 (0.54)      | 2.84 (0.68)          | 0.000***   |
| Agreeableness                      | 3.76 (0.49)      | 3.60 (0.50)          | 0.117      |
| Conscientiousness                  | 3.56 (0.54)      | 3.11 (0.64)          | 0.001***   |
| Neuroticism                        | 2.65 (0.55)      | 3.13 (0.55)          | 0.000***   |
| Openness                           | 3.27 (0.61)      | 3.45 (0.71)          | 0.168      |

Notes: Table reports means and standard deviations (in parentheses), or proportions. The  $p$ -values correspond to Welch's two-sample  $t$ -tests for differences between the Substance Use Disorder (SUD) group and non-dependent Control group. Significance levels: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table 6 presents summary statistics on substance use history, primary drug classifications, initiation circumstances, family structure, and relapse factors for the SUD clinical sample. Due to institutional constraints at the compulsory rehabilitation facilities, data collection regarding participants' background was restricted to these specific points.

Table 6: Summary of survey responses for persons with SUD

| Variable                                                          | SUD Sample   | Variable                                                | SUD Sample |
|-------------------------------------------------------------------|--------------|---------------------------------------------------------|------------|
| Age at first substance use (years)                                | 23.64 (7.56) | <b>Family structure (Multiple choices allowed)</b>      |            |
| <b>Substance use frequency</b>                                    |              | Mother                                                  | 76.40%     |
| Once every few days                                               | 62.12%       | Father                                                  | 66.29%     |
| Once a day                                                        | 15.15%       | Children                                                | 43.82%     |
| Multiple times a day                                              | 22.73%       | Grandparents                                            | 35.96%     |
| <b>Primary substance classification</b>                           |              | Spouse / Partner                                        | 21.35%     |
| Synthetic substances only (e.g., Methamphetamine, Ketamine)       | 58.21%       | None (Living alone / No family)                         | 1.12%      |
| Traditional substances only (e.g., Heroin, Opium)                 | 22.39%       | <b>Relapse history and factors</b>                      |            |
| Mainly synthetic substances                                       | 14.93%       | <b>Prior relapses / detentions</b>                      |            |
| Mainly traditional substances                                     | 7.46%        | 1 time                                                  | 66.67%     |
| <b>Reasons for first substance use (Multiple choices allowed)</b> |              | 2 times                                                 | 27.27%     |
| Curiosity                                                         | 58.21%       | 3+ times                                                | 6.06%      |
| Seeking thrills / Pleasure                                        | 40.30%       | <b>Duration of abstinence prior to relapse</b>          |            |
| Peer pressure / Temptation                                        | 22.39%       | < 1 month                                               | 17.39%     |
| Relieving stress / Distress                                       | 11.94%       | 1–3 months                                              | 13.04%     |
| Weight loss                                                       | 7.46%        | 4–6 months                                              | 8.70%      |
| Staying awake for gaming                                          | 4.48%        | 7–12 months                                             | 21.74%     |
| Socializing / Making friends                                      | 2.99%        | 1–3 years                                               | 21.74%     |
| Influence of family members                                       | 2.99%        | > 3 years                                               | 17.39%     |
| Medical treatment / Illness                                       | 0.00%        | <b>Reasons for relapse (Multiple choices allowed)</b>   |            |
| Deceived / Drugged by others                                      | 0.00%        | Psychological craving (Seeking sensation of substances) | 48.15%     |
| Other reasons                                                     | 2.99%        | Physical discomfort / Withdrawal                        | 22.22%     |
| <b>Location of first substance use</b>                            |              | Unemployment / Idleness                                 | 22.22%     |
| Relative or friend's home                                         | 34.33%       | Temptation from peers                                   | 18.52%     |
| Own home                                                          | 25.37%       | Social discrimination / Stigma                          | 14.81%     |
| Bar, nightclub, or KTV                                            | 16.42%       | Family conflict / Lack of affection                     | 7.41%      |
| Hotel / Lodging                                                   | 16.42%       |                                                         |            |
| Video game arcade                                                 | 4.48%        |                                                         |            |
| Restaurant                                                        | 1.49%        |                                                         |            |
| Workplace                                                         | 1.49%        |                                                         |            |
| School                                                            | 1.49%        |                                                         |            |
| Gambling venue                                                    | 1.49%        |                                                         |            |
| Other locations                                                   | 1.49%        |                                                         |            |

*Notes:* Table reports summary statistics for the Substance Use Disorder (SUD) group ( $N = 95$ ). Continuous variables report the sample mean and standard deviation in parentheses; categorical and multi-select items report percentage shares among respondents answering the respective question.

## F Other Rationality Indices & Robustness Checks

To ensure our findings are robust rationality index, we redo the analysis for the Average Varian Index (AVI), the Houtman-Maks Index (HMI), and the Minimum Cost Index ( $1 - \text{MCI}$ ). Across all alternative metrics, the results are fully consistent with the main CCEI findings reported in the paper.

While CCEI applies a uniform efficiency scalar to all budgets, Varian's Index applies an observation-specific perturbation  $v_t \in [0, 1]$  to the budget of each trial  $t$ . A bundle  $\mathbf{q}_i$  is revealed preferred to  $\mathbf{q}_j$  under perturbation vector  $\mathbf{v}$  if  $v_i \mathbf{p}_i \cdot \mathbf{q}_i \geq \mathbf{p}_i \cdot \mathbf{q}_j$ .

The AVI seeks the minimal distance from the unperturbed state ( $\mathbf{v} = 1$ ) such that the data satisfies  $\mathbf{v}$ -GARP:

$$\text{AVI} = \min_{\mathbf{v} \in [0,1]^T} \frac{1}{T} \sum_{t=1}^T (1 - v_t) \quad \text{subject to } \mathbf{v}\text{-GARP} \quad (16)$$

Table 7 evaluates rationality using Varian's budget perturbation metric (AVI).

The HMI counts discrete observations rather than scaling budget sets. It is defined as the size of the largest subset of the subject's choices that is mutually consistent with GARP (Houtman and Maks, 1985):

$$\text{HMI} = \max_{S \subseteq T} \{|S| : \text{The observations in } S \text{ satisfy GARP}\} \quad (17)$$

The HMI provides a cardinal count of consistent decisions, making it less sensitive to the magnitude of budget violations than the CCEI. Table 8 evaluates rationality using the Houtman-Maks Index (HMI).

The MCI measures the monetary cost required to break all cyclical revealed preference relations. Let  $R_0$  be the set of direct revealed preference relations ( $q_i R_0 q_j \iff \mathbf{p}_i \cdot \mathbf{q}_i \geq \mathbf{p}_i \cdot \mathbf{q}_j$ ). The monetary cost of removing relation  $(i, j)$  is the surplus expenditure  $c_{ij} = \mathbf{p}_i \cdot (\mathbf{q}_i - \mathbf{q}_j)$ .

The MCI finds the subset of relations  $B \subset R_0$  whose removal makes the remaining relation acyclic, minimizing the total cost normalized by total expenditure:

$$\text{MCI} = \min_{B \subset R_0 : R_0 \setminus B \text{ is acyclic}} \frac{\sum_{(i,j) \in B} \mathbf{p}_i \cdot (\mathbf{q}_i - \mathbf{q}_j)}{\sum_{t=1}^T \mathbf{p}_t \cdot \mathbf{q}_t} \quad (18)$$

Table 9 evaluates rationality using the Minimum Cost Index ( $1 - \text{MCI}$ ).

Table 7: AVI Robustness

| Dependent Variable                           | (i) Without Controls |          |            |          | (ii) With Controls |          |            |          |
|----------------------------------------------|----------------------|----------|------------|----------|--------------------|----------|------------|----------|
|                                              | Coeff.               | SE       | Unadj. $p$ | FWER $p$ | Coeff.             | SE       | Unadj. $p$ | FWER $p$ |
| <b>Panel A: Ordinary Least Squares (OLS)</b> |                      |          |            |          |                    |          |            |          |
| AVI (Risk)                                   | -0.0125*             | (0.0047) | 0.023      | 0.064    | -0.0178            | (0.0073) | 0.042      | 0.111    |
| AVI (Time)                                   | -0.0088***           | (0.0023) | 0.004      | 0.010    | -0.0138            | (0.0056) | 0.068      | 0.111    |
| AVI (Social)                                 | 0.0012               | (0.0039) | 0.791      | 0.791    | -0.0006            | (0.0053) | 0.929      | 0.929    |
| AVI FOSD (Risk)                              | -0.0402***           | (0.0072) | 0.001      | 0.001    | -0.0487**          | (0.0120) | 0.002      | 0.024    |
| FOSD Compliance (Risk)                       | -0.1380              | (0.0283) | 0.001      | 0.103    | -0.1580            | (0.0421) | 0.004      | 0.191    |
| <b>Panel B: Tobit Regressions</b>            |                      |          |            |          |                    |          |            |          |
| AVI (Risk)                                   | -0.0191              | (0.0081) | 0.085      | 0.109    | -0.0246*           | (0.0095) | 0.068      | 0.082    |
| AVI (Time)                                   | -0.0176**            | (0.0051) | 0.009      | 0.028    | -0.0274*           | (0.0092) | 0.049      | 0.077    |
| AVI (Social)                                 | 0.0005               | (0.0060) | 0.938      | 0.938    | -0.0032            | (0.0076) | 0.725      | 0.725    |
| AVI FOSD (Risk)                              | -0.0426***           | (0.0083) | 0.001      | 0.001    | -0.0525**          | (0.0124) | 0.003      | 0.023    |
| FOSD Compliance (Risk)                       | -0.1507              | (0.0322) | 0.001      | 0.108    | -0.1693            | (0.0427) | 0.005      | 0.307    |

Notes: Table reports regression coefficients for the Substance Use Disorder (SUD) indicator (SUD = 1, Control = 0). Conditional specifications control for gender, education, income, age, and Big Five personality traits. The sample consists of  $N = 129$  participants for OLS models and  $N = 129$  participants for Tobit regressions. Robust standard errors are reported in parentheses for OLS models; standard errors for Tobit regressions. FWER-adjusted  $p$ -values are computed via Romano-Wolf stepdown resampling with 1,000 permutations over the variables in each panel. Significance stars are based on FWER-adjusted  $p$ -values: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table 8: HMI Robustness

| Dependent Variable                           | (i) Without Controls |          |            |          | (ii) With Controls |          |            |          |
|----------------------------------------------|----------------------|----------|------------|----------|--------------------|----------|------------|----------|
|                                              | Coeff.               | SE       | Unadj. $p$ | FWER $p$ | Coeff.             | SE       | Unadj. $p$ | FWER $p$ |
| <b>Panel A: Ordinary Least Squares (OLS)</b> |                      |          |            |          |                    |          |            |          |
| HMI (Risk)                                   | -0.0656***           | (0.0177) | 0.007      | 0.008    | -0.0914***         | (0.0218) | 0.001      | 0.003    |
| HMI (Time)                                   | -0.0468***           | (0.0133) | 0.001      | 0.008    | -0.0632***         | (0.0203) | 0.007      | 0.008    |
| HMI (Social)                                 | -0.0040              | (0.0170) | 0.824      | 0.824    | -0.0144            | (0.0247) | 0.600      | 0.600    |
| HMI FOSD (Risk)                              | -0.1305***           | (0.0204) | 0.001      | 0.001    | -0.1435***         | (0.0267) | 0.001      | 0.001    |
| FOSD Compliance (Risk)                       | -0.1380              | (0.0283) | 0.001      | 0.103    | -0.1580            | (0.0421) | 0.004      | 0.191    |
| <b>Panel B: Tobit Regressions</b>            |                      |          |            |          |                    |          |            |          |
| HMI (Risk)                                   | -0.0810***           | (0.0230) | 0.003      | 0.003    | -0.1118***         | (0.0262) | 0.001      | 0.003    |
| HMI (Time)                                   | -0.0675***           | (0.0205) | 0.001      | 0.003    | -0.0914***         | (0.0271) | 0.004      | 0.007    |
| HMI (Social)                                 | 0.0044               | (0.0248) | 0.874      | 0.874    | -0.0169            | (0.0338) | 0.684      | 0.684    |
| HMI FOSD (Risk)                              | -0.1307***           | (0.0209) | 0.001      | 0.001    | -0.1433***         | (0.0251) | 0.001      | 0.001    |
| FOSD Compliance (Risk)                       | -0.1507              | (0.0322) | 0.001      | 0.108    | -0.1693            | (0.0427) | 0.005      | 0.307    |

Notes: Table reports regression coefficients for the Substance Use Disorder (SUD) indicator (SUD = 1, Control = 0). Conditional specifications control for gender, education, income, age, and Big Five personality traits. The sample consists of  $N = 129$  participants for OLS models and  $N = 129$  participants for Tobit regressions. Robust standard errors are reported in parentheses for OLS models; standard errors for Tobit regressions. FWER-adjusted  $p$ -values are computed via Romano-Wolf stepdown resampling with 1,000 permutations over the variables in each panel. Significance stars are based on FWER-adjusted  $p$ -values: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table 9: 1 – MCI Robustness

| Dependent Variable                           | (i) Without Controls |          |            |          | (ii) With Controls |          |            |          |
|----------------------------------------------|----------------------|----------|------------|----------|--------------------|----------|------------|----------|
|                                              | Coeff.               | SE       | Unadj. $p$ | FWER $p$ | Coeff.             | SE       | Unadj. $p$ | FWER $p$ |
| <b>Panel A: Ordinary Least Squares (OLS)</b> |                      |          |            |          |                    |          |            |          |
| 1 – MCI (Risk)                               | -0.0443              | (0.0216) | 0.165      | 0.239    | -0.0532            | (0.0305) | 0.379      | 0.429    |
| 1 – MCI (Time)                               | -0.0190              | (0.0060) | 0.009      | 0.162    | -0.0297            | (0.0134) | 0.081      | 0.384    |
| 1 – MCI (Social)                             | 0.0023               | (0.0099) | 0.819      | 0.819    | -0.0030            | (0.0136) | 0.834      | 0.834    |
| 1 – MCI FOSD (Risk)                          | -0.1880              | (0.0489) | 0.017      | 0.135    | -0.2343            | (0.0769) | 0.059      | 0.292    |
| FOSD Compliance (Risk)                       | -0.1380              | (0.0283) | 0.001      | 0.103    | -0.1580            | (0.0421) | 0.004      | 0.191    |
| <b>Panel B: Tobit Regressions</b>            |                      |          |            |          |                    |          |            |          |
| 1 – MCI (Risk)                               | -0.0612              | (0.0278) | 0.150      | 0.174    | -0.0836            | (0.0438) | 0.397      | 0.434    |
| 1 – MCI (Time)                               | -0.0421              | (0.0138) | 0.017      | 0.146    | -0.0640            | (0.0231) | 0.060      | 0.382    |
| 1 – MCI (Social)                             | 0.0005               | (0.0150) | 0.979      | 0.979    | -0.0103            | (0.0195) | 0.671      | 0.671    |
| 1 – MCI FOSD (Risk)                          | -0.1584*             | (0.0312) | 0.001      | 0.095    | -0.2212            | (0.0579) | 0.011      | 0.321    |
| FOSD Compliance (Risk)                       | -0.1507              | (0.0322) | 0.001      | 0.108    | -0.1693            | (0.0427) | 0.005      | 0.307    |

Notes: Table reports regression coefficients for the Substance Use Disorder (SUD) indicator (SUD = 1, Control = 0). Conditional specifications control for gender, education, income, age, and Big Five personality traits. The sample consists of  $N = 129$  participants for OLS models and  $N = 129$  participants for Tobit regressions. Robust standard errors are reported in parentheses for OLS models; standard errors for Tobit regressions. FWER-adjusted  $p$ -values are computed via Romano-Wolf stepdown resampling with 1,000 permutations over the variables in each panel. Significance stars are based on FWER-adjusted  $p$ -values: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

## G Secondary Structural Preference Parameters

This section presents regression results for the secondary structural preference parameters: risk utility curvature ( $\rho_r$ ), intertemporal consumption curvature ( $\rho_t$ ), and altruistic substitution curvature ( $\rho_s$ ).

Figure 8 visualizes the outlier structure for these parameters using standard boxplots.

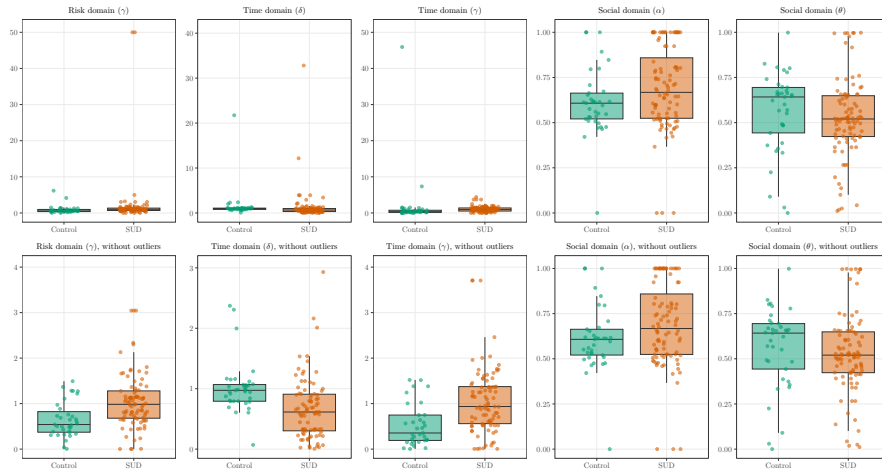


Figure 8: Boxplots displaying the distribution and extreme outliers of the structural parameters.

## H Detailed Regression Models with All Control Variable Coefficients

For completeness, this section provides the full detailed regression tables corresponding to Tables 1–4 in the main paper, displaying all individual control variable coefficients for demographic and Big

Five personality covariates. The reference omitted categories for categorical controls are Male (Gender), Unspecified (Education), and Unspecified (Income).

Table 10: Full Regression Controls – CCEI &amp; Revealed Preference Indices (OLS)

| Independent Variable           | Risk CCEI              | Time CCEI              | Social CCEI           | Symmetric Risk         | FOSD Compliance        |
|--------------------------------|------------------------|------------------------|-----------------------|------------------------|------------------------|
| Constant                       | 1.2517***<br>(0.2308)  | 0.8425***<br>(0.2091)  | 0.8875***<br>(0.1746) | 1.3811***<br>(0.3984)  | 1.1773***<br>(0.2480)  |
| <b>SUD Indicator (SUD = 1)</b> | -0.1308***<br>(0.0395) | -0.0911**<br>(0.0385)  | -0.0059<br>(0.0307)   | -0.2217***<br>(0.0595) | -0.1580***<br>(0.0421) |
| Age (Years)                    | 0.0005<br>(0.0013)     | 0.0006<br>(0.0011)     | -0.0009<br>(0.0018)   | 0.0014<br>(0.0021)     | 0.0002<br>(0.0014)     |
| Gender: Female                 | -0.0403<br>(0.0283)    | -0.0349<br>(0.0266)    | -0.0136<br>(0.0300)   | -0.0567<br>(0.0542)    | -0.0187<br>(0.0306)    |
| Edu: High School or Below      | 0.0044<br>(0.0370)     | 0.0016<br>(0.0372)     | -0.0116<br>(0.0302)   | 0.0783<br>(0.0626)     | 0.0644*<br>(0.0357)    |
| Edu: Master's+                 | -0.0961*<br>(0.0503)   | -0.0377<br>(0.0265)    | -0.0802*<br>(0.0430)  | -0.1052<br>(0.0970)    | -0.0147<br>(0.0562)    |
| Income: < 100,000 RMB          | -0.1802***<br>(0.0564) | -0.1378***<br>(0.0476) | -0.0621<br>(0.0525)   | -0.3767***<br>(0.0876) | -0.2318***<br>(0.0512) |
| Income: 100,000–500,000 RMB    | -0.2421***<br>(0.0728) | -0.1713**<br>(0.0668)  | -0.0818**<br>(0.0387) | -0.4581***<br>(0.0985) | -0.2423***<br>(0.0636) |
| Income: > 500,000 RMB          | -0.1090*<br>(0.0568)   | -0.1019**<br>(0.0425)  | -0.0283<br>(0.0526)   | -0.2901**<br>(0.1135)  | -0.1609**<br>(0.0647)  |
| Big 5: Extraversion            | 0.0071<br>(0.0212)     | 0.0009<br>(0.0210)     | -0.0273<br>(0.0274)   | -0.0245<br>(0.0464)    | 0.0116<br>(0.0314)     |
| Big 5: Agreeableness           | 0.0566*<br>(0.0332)    | 0.0533*<br>(0.0279)    | 0.0550*<br>(0.0285)   | -0.0060<br>(0.0585)    | -0.0060<br>(0.0403)    |
| Big 5: Conscientiousness       | -0.0268<br>(0.0396)    | 0.0188<br>(0.0336)     | 0.0220<br>(0.0323)    | 0.0190<br>(0.0716)     | 0.0101<br>(0.0394)     |
| Big 5: Neuroticism             | -0.0285<br>(0.0329)    | 0.0134<br>(0.0266)     | 0.0106<br>(0.0233)    | -0.0129<br>(0.0579)    | -0.0265<br>(0.0347)    |
| Big 5: Openness                | -0.0435*<br>(0.0260)   | -0.0063<br>(0.0252)    | -0.0184<br>(0.0166)   | -0.0299<br>(0.0439)    | -0.0157<br>(0.0286)    |
| Observations                   | 103                    | 103                    | 103                   | 103                    | 103                    |
| $R^2$                          | 0.196                  | 0.121                  | 0.086                 | 0.188                  | 0.208                  |

Notes: Table reports full covariate coefficients. Table reports OLS regressions on the trimmed sample. Reference categories: Male (Gender), Unspecified (Education), Unspecified (Income). Significance levels: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table 11: Full Regression Controls – CCEI &amp; Revealed Preference Indices (Tobit)

| Independent Variable           | Risk CCEI              | Time CCEI              | Social CCEI            | Symmetric Risk         | FOSD Compliance        |
|--------------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| Constant                       | 2.0194***<br>(0.3127)  | 1.8209***<br>(0.3556)  | 1.6367***<br>(0.3058)  | 1.2847***<br>(0.3805)  | 1.1737***<br>(0.2440)  |
| <b>SUD Indicator (SUD = 1)</b> | -0.1660***<br>(0.0471) | -0.1668***<br>(0.0600) | -0.0142<br>(0.0452)    | -0.2213***<br>(0.0556) | -0.1693***<br>(0.0427) |
| Age (Years)                    | 0.0006<br>(0.0016)     | 0.0004<br>(0.0017)     | -0.0010<br>(0.0024)    | 0.0018<br>(0.0021)     | 0.0002<br>(0.0014)     |
| Gender: Female                 | -0.0492<br>(0.0352)    | -0.0684*<br>(0.0377)   | -0.0514<br>(0.0423)    | -0.0583<br>(0.0510)    | -0.0180<br>(0.0308)    |
| Edu: High School or Below      | 0.1606**<br>(0.0683)   | 0.1277*<br>(0.0751)    | 0.1510**<br>(0.0749)   | 0.1864*<br>(0.1004)    | 0.0885<br>(0.0610)     |
| Edu: Bachelor's or College     | 0.1370***<br>(0.0531)  | 0.0909**<br>(0.0433)   | 0.1663***<br>(0.0609)  | 0.1065<br>(0.0911)     | 0.0230<br>(0.0539)     |
| Edu: Master's+                 | NA<br>(0.0000)         | NA<br>(0.0000)         | NA<br>(0.0000)         | NA<br>(0.0000)         | NA<br>(0.0000)         |
| Income: < 100,000 RMB          | -1.0870***<br>(0.1408) | -1.1061***<br>(0.1848) | -0.9893***<br>(0.1537) | -0.3857***<br>(0.0833) | -0.2260***<br>(0.0523) |
| Income: 100,000–500,000 RMB    | -1.1585***<br>(0.1602) | -1.1394***<br>(0.2142) | -1.0279***<br>(0.1552) | -0.4580***<br>(0.0928) | -0.2284***<br>(0.0664) |
| Income: > 500,000 RMB          | -0.9746***<br>(0.1293) | -1.0227***<br>(0.1698) | -0.8684***<br>(0.1449) | -0.2767**<br>(0.1130)  | -0.1397**<br>(0.0688)  |
| Big 5: Extraversion            | -0.0055<br>(0.0300)    | -0.0018<br>(0.0362)    | -0.0600<br>(0.0444)    | -0.0215<br>(0.0442)    | 0.0098<br>(0.0319)     |
| Big 5: Agreeableness           | 0.0851**<br>(0.0418)   | 0.0720<br>(0.0457)     | 0.0970**<br>(0.0494)   | -0.0068<br>(0.0559)    | -0.0060<br>(0.0392)    |
| Big 5: Conscientiousness       | -0.0243<br>(0.0456)    | 0.0429<br>(0.0456)     | 0.0406<br>(0.0449)     | 0.0151<br>(0.0672)     | 0.0147<br>(0.0401)     |
| Big 5: Neuroticism             | -0.0289<br>(0.0390)    | 0.0061<br>(0.0427)     | 0.0234<br>(0.0448)     | -0.0116<br>(0.0536)    | -0.0265<br>(0.0340)    |
| Big 5: Openness                | -0.0492<br>(0.0319)    | -0.0378<br>(0.0378)    | -0.0324<br>(0.0292)    | -0.0338<br>(0.0418)    | -0.0234<br>(0.0296)    |
| Observations                   | 103                    | 103                    | 103                    | 103                    | 103                    |
| Pseudo $R^2$                   | -61.926                | 0.628                  | 0.496                  | 1.196                  | -0.492                 |

Notes: Table reports full covariate coefficients. Table reports Tobit regressions on the full sample. Reference categories: Male (Gender), Unspecified (Education), Unspecified (Income). Significance levels: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . The Master's+ education dummy is dropped (NA) due to perfect collinearity/complete separation during maximum likelihood estimation caused by its sparse sample size ( $N = 13$ ).

Table 12: Full Regression Controls – Structural Preference Parameters (OLS)

| Independent Variable           | Risk Wt ( $\alpha_r$ ) | Risk CRRA ( $\rho_r$ ) | Time Disc ( $\delta$ ) | Time CRRA ( $\rho_t$ ) | Social Wt ( $\alpha_s$ ) | Social CES ( $\rho_s$ ) |
|--------------------------------|------------------------|------------------------|------------------------|------------------------|--------------------------|-------------------------|
| Constant                       | 0.5397***<br>(0.1735)  | 1.5909**<br>(0.7064)   | 2.2023***<br>(0.6803)  | -1.2792*<br>(0.7300)   | 0.2438<br>(0.2272)       | 0.1739<br>(0.7493)      |
| <b>SUD Indicator (SUD = 1)</b> | 0.0207<br>(0.0372)     | 0.3159**<br>(0.1332)   | -0.4917***<br>(0.1049) | 0.4130***<br>(0.1248)  | 0.0650<br>(0.0421)       | 0.0957<br>(0.1480)      |
| Age (Years)                    | 0.0018<br>(0.0019)     | -0.0071<br>(0.0053)    | -0.0028<br>(0.0057)    | -0.0049<br>(0.0056)    | 0.0037*<br>(0.0022)      | 0.0005<br>(0.0061)      |
| Gender: Female                 | 0.0394<br>(0.0347)     | 0.1811<br>(0.1103)     | -0.0018<br>(0.1475)    | -0.0539<br>(0.1346)    | 0.0167<br>(0.0428)       | -0.0784<br>(0.1331)     |
| Edu: High School or Below      | -0.0119<br>(0.0324)    | -0.0522<br>(0.1619)    | -0.0503<br>(0.1795)    | -0.0373<br>(0.1758)    | -0.0070<br>(0.0593)      | -0.0321<br>(0.1728)     |
| Edu: Master's+                 | 0.0839<br>(0.0641)     | 0.1047<br>(0.1573)     | -0.2334*<br>(0.1235)   | 0.0681<br>(0.1839)     | 0.0586<br>(0.0920)       | 0.0421<br>(0.2528)      |
| Income: < 100,000 RMB          | -0.0365<br>(0.0559)    | 0.1134<br>(0.2001)     | -0.1341<br>(0.2193)    | 1.1804***<br>(0.2027)  | 0.0939<br>(0.0626)       | -0.1986<br>(0.2249)     |
| Income: 100,000–500,000 RMB    | -0.0010<br>(0.0462)    | 0.0651<br>(0.2165)     | -0.3940*<br>(0.2256)   | 1.0964***<br>(0.2232)  | 0.1451**<br>(0.0678)     | -0.0606<br>(0.2345)     |
| Income: > 500,000 RMB          | -0.0477<br>(0.0593)    | -0.0475<br>(0.2859)    | -0.2224<br>(0.2454)    | 0.6656**<br>(0.2575)   | 0.2198**<br>(0.0862)     | 0.1822<br>(0.2798)      |
| Big 5: Extraversion            | -0.0035<br>(0.0338)    | -0.0484<br>(0.1335)    | -0.0226<br>(0.1014)    | 0.0225<br>(0.1029)     | 0.0231<br>(0.0341)       | -0.0570<br>(0.1653)     |
| Big 5: Agreeableness           | 0.0081<br>(0.0348)     | -0.2052*<br>(0.1202)   | 0.0995<br>(0.1242)     | -0.1213<br>(0.1239)    | -0.0241<br>(0.0438)      | 0.0789<br>(0.1405)      |
| Big 5: Conscientiousness       | -0.0383<br>(0.0343)    | -0.0156<br>(0.1559)    | -0.0457<br>(0.0985)    | 0.2145*<br>(0.1172)    | -0.0185<br>(0.0366)      | -0.0832<br>(0.2030)     |
| Big 5: Neuroticism             | -0.0042<br>(0.0271)    | -0.0266<br>(0.1080)    | -0.2681**<br>(0.1131)  | 0.0123<br>(0.1128)     | 0.0378<br>(0.0381)       | 0.0810<br>(0.1254)      |
| Big 5: Openness                | 0.0125<br>(0.0178)     | 0.0677<br>(0.0831)     | -0.0315<br>(0.0966)    | 0.1442<br>(0.1053)     | 0.0217<br>(0.0333)       | -0.0013<br>(0.0943)     |
| Observations                   | 103                    | 96                     | 98                     | 98                     | 103                      | 97                      |
| $R^2$                          | 0.094                  | 0.107                  | 0.211                  | 0.242                  | 0.125                    | 0.074                   |

Notes: Table reports full covariate coefficients. Table reports OLS regressions on the trimmed sample. Reference categories: Male (Gender), Unspecified (Education), Unspecified (Income). Significance levels: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table 13: Full Regression Controls – Structural Preference Parameters (Median)

| Independent Variable           | Risk Wt ( $\alpha_r$ ) | Risk CRRA ( $\rho_r$ ) | Time Disc ( $\delta$ ) | Time CRRA ( $\rho_t$ ) | Social Wt ( $\alpha_s$ ) | Social CES ( $\rho_s$ ) |
|--------------------------------|------------------------|------------------------|------------------------|------------------------|--------------------------|-------------------------|
| Constant                       | 0.4986*<br>(0.2606)    | 1.1334<br>(1.0536)     | 2.2172**<br>(0.8471)   | -1.7141<br>(1.3545)    | 0.1175<br>(0.3737)       | 0.5094<br>(1.1530)      |
| <b>SUD Indicator (SUD = 1)</b> | -0.0074<br>(0.0297)    | 0.4004*<br>(0.2120)    | -0.5028***<br>(0.1405) | 0.4956*<br>(0.2558)    | 0.1142*<br>(0.0613)      | -0.1286<br>(0.2135)     |
| Age (Years)                    | 0.0001<br>(0.0013)     | -0.0004<br>(0.0086)    | 0.0004<br>(0.0049)     | -0.0038<br>(0.0091)    | 0.0033<br>(0.0031)       | -0.0053<br>(0.0093)     |
| Gender: Female                 | 0.0399<br>(0.0285)     | 0.2007<br>(0.1674)     | 0.2415*<br>(0.1393)    | -0.1229<br>(0.2729)    | 0.0101<br>(0.0581)       | -0.1065<br>(0.1807)     |
| Edu: High School or Below      | 0.0081<br>(0.0286)     | -0.2934<br>(0.2047)    | -0.1747<br>(0.1442)    | 0.0494<br>(0.3504)     | 0.0404<br>(0.0629)       | -0.0474<br>(0.1954)     |
| Edu: Master's+                 | 0.0387<br>(0.0833)     | 0.3761<br>(0.2849)     | -0.1788<br>(0.1449)    | 0.2474<br>(0.3385)     | 0.1327<br>(0.1175)       | 0.1480<br>(0.8284)      |
| Income: < 100,000 RMB          | -0.0141<br>(0.1953)    | 0.0095<br>(0.5855)     | -0.0669<br>(0.6463)    | 0.9606<br>(0.7920)     | 0.1313<br>(0.2084)       | -0.2746<br>(0.6001)     |
| Income: 100,000–500,000 RMB    | 0.0030<br>(0.1980)     | 0.0186<br>(0.6073)     | -0.2271<br>(0.6648)    | 0.9243<br>(0.8442)     | 0.2407<br>(0.2138)       | -0.3033<br>(0.6146)     |
| Income: > 500,000 RMB          | -0.0179<br>(0.1947)    | -0.1594<br>(0.6586)    | 0.1086<br>(0.6398)     | 0.3771<br>(0.8764)     | 0.1690<br>(0.2305)       | -0.1730<br>(0.6630)     |
| Big 5: Extraversion            | -0.0068<br>(0.0227)    | -0.0915<br>(0.1499)    | -0.0995<br>(0.1212)    | 0.0722<br>(0.2009)     | 0.0081<br>(0.0393)       | 0.0625<br>(0.2102)      |
| Big 5: Agreeableness           | -0.0130<br>(0.0285)    | -0.3031*<br>(0.1665)   | 0.0058<br>(0.1156)     | 0.0848<br>(0.2199)     | -0.0597<br>(0.0649)      | 0.1340<br>(0.1614)      |
| Big 5: Conscientiousness       | 0.0021<br>(0.0296)     | 0.0263<br>(0.1073)     | 0.0057<br>(0.1336)     | 0.1193<br>(0.2197)     | 0.0424<br>(0.0444)       | -0.2488<br>(0.2480)     |
| Big 5: Neuroticism             | -0.0060<br>(0.0269)    | 0.0842<br>(0.1170)     | -0.2583**<br>(0.1005)  | 0.0856<br>(0.2022)     | 0.0587<br>(0.0626)       | 0.1024<br>(0.1957)      |
| Big 5: Openness                | 0.0193<br>(0.0233)     | 0.1581<br>(0.1098)     | -0.0199<br>(0.1043)    | 0.0635<br>(0.1830)     | 0.0084<br>(0.0490)       | 0.0233<br>(0.1363)      |
| Observations                   | 103                    | 103                    | 103                    | 103                    | 103                      | 103                     |
| Pseudo $R^2$                   | 0.049                  | 0.041                  | 0.067                  | 0.075                  | 0.097                    | 0.027                   |

Notes: Table reports full covariate coefficients. Table reports Median regressions on the full sample. Reference categories: Male (Gender), Unspecified (Education), Unspecified (Income). Significance levels: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .